SAECOIN — MINING GUIDE

Document: Mining_Guide.md
Version: 1.0
Project: SAECOIN
Ticker: SAC (Provisional)
Consensus: Proof of Work (PoW)
Transaction Model: UTXO
Status: Draft / Planning

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1. PURPOSE

This document defines the proposed mining architecture and operational requirements for SAECOIN.

The goal is to support decentralized mining participation through:

- CPU Mining
- GPU Mining
- Solo Mining
- Pool Mining (Future)
- Mining Performance Monitoring
- Mining Reward Tracking

SAECOIN aims to make mining accessible to ordinary users while maintaining network security.

CPU and GPU mining support will depend on the final Proof of Work algorithm.

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2. WHAT IS SAECOIN MINING?

Mining is the process of searching for a valid Proof of Work solution.

Miners:

1. Obtain a candidate block template.
2. Receive valid transaction information.
3. Construct or receive a candidate block.
4. Search for a valid nonce or other algorithm-specific solution.
5. Submit a candidate solution.
6. The node validates the block.
7. A valid block is propagated to the network.

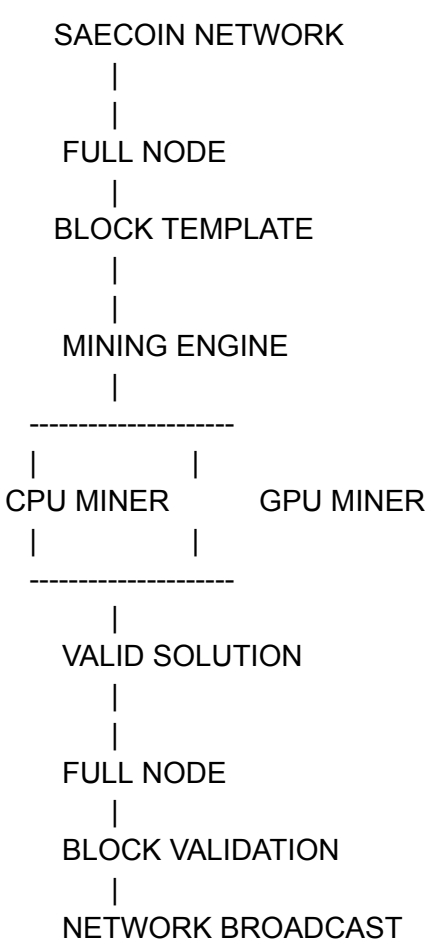
The first valid block accepted into the best-work chain may earn the applicable block subsidy and transaction fees.

Mining does not guarantee income.

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3. MINING ARCHITECTURE

Proposed architecture:



The mining engine should be separated from the user interface.

The node must independently validate every submitted solution.

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4. PROOF OF WORK ALGORITHM

The final PoW algorithm has not been selected.

Algorithm selection must consider:

- Security
- CPU performance
- GPU performance
- Memory requirements
- Verification speed
- Implementation complexity
- Mining decentralization
- ASIC resistance objectives
- Hardware accessibility

The algorithm must be reviewed and benchmarked before implementation.

No mining software should assume an unfinished algorithm.

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5. CPU MINING

CPU mining should allow users to participate using ordinary processors.

Proposed CPU miner features:

- Automatic CPU detection
- Configurable thread count
- CPU usage control
- Hashrate monitoring
- Mining start/stop
- Node connection
- Mining status
- Error reporting

Example future command:

saecoin-miner.exe --mode cpu --threads 4

This is a proposed command interface, not an implemented feature.

5.1 CPU Mining Configuration

Users should be able to configure:

- Number of CPU threads
- Maximum CPU usage
- Mining destination address
- Node connection
- Mining intensity

The miner should avoid consuming all available CPU resources by default.

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6. GPU MINING

GPU mining should allow supported graphics cards to participate.

Potential development technologies include:

- CUDA
- OpenCL
- Other suitable GPU computing frameworks

The final technology depends on the selected PoW algorithm and supported hardware.

Proposed GPU miner features:

- GPU detection
- Device selection
- Hashrate monitoring
- Temperature monitoring, where available
- GPU utilization monitoring
- Mining start/stop
- Error reporting
- Multiple GPU support, if practical

Example future command:

saecoin-miner.exe --mode gpu

This is illustrative only.

GPU mining support must be tested on supported hardware before release.

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7. CPU AND GPU COMPATIBILITY

The project aims to support both CPU and GPU miners.

The final algorithm should be evaluated against:

Requirement| Objective
CPU mining| Supported
GPU mining| Supported
Multi-core CPU| Supported where practical
Multiple GPUs| Evaluated
Memory efficiency| Important

Verification speed| Important
ASIC resistance| Design objective

CPU and GPU miners must produce solutions that satisfy exactly the same consensus rules.

Different miner implementations must not create different block-validity rules.

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8. SOLO MINING

Solo mining means mining independently through a user's own Full Node.

Proposed process:

1. Install SAECOIN Full Node.
2. Synchronize with the network.
3. Create or import a mining wallet.
4. Configure the mining destination address.
5. Start the miner.
6. Obtain block templates from the node.
7. Search for valid Proof of Work.
8. Submit a valid solution.
9. Node validates the block.
10. Valid blocks are broadcast to the network.

Solo mining rewards are not guaranteed.

A miner may spend considerable time without finding a block.

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9. POOL MINING

Pool mining allows multiple miners to contribute computing power toward a shared mining operation.

The pool coordinates mining work and distributes rewards according to an agreed accounting method.

Proposed pool components:

SAECOIN MINING POOL
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|-- Pool Server
|
|-- Full Node
|

|-- Work Distribution
|
|-- Share Verification
|
|-- Miner Accounting
|
|-- Reward Distribution
|
|-- Monitoring Dashboard

Pool mining is a planned future feature.

It should not delay the initial implementation of solo mining.

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10. MINING POOL REQUIREMENTS

A future mining pool should support:

- Miner registration
- Worker identification
- Job distribution
- Share submission
- Share validation
- Invalid share rejection
- Hashrate calculation
- Miner statistics
- Reward accounting
- Payout records
- Pool health monitoring

The pool must never be able to create invalid blocks.

The Full Node must independently validate all blocks.

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11. MINING REWARD

Proposed monetary parameters:

Parameter| Proposed Value
Initial block reward| 50 SAC
Maximum supply target| 21,000,000 SAC
Halving interval| 210,000 blocks
Premine| 0 SAC
Decimal precision| 8

All values are provisional.

The final issuance schedule must be mathematically verified using integer arithmetic.

The actual total issuance must never exceed the finalized maximum supply.

Transaction fees may also be included in miner rewards according to the final consensus specification.

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12. BLOCK REWARD MATURITY

A newly mined block may have a maturity period before its reward becomes spendable.

The following rules must be finalized:

- Coinbase transaction structure
- Coinbase maturity period
- Reward validation
- Reward unlock conditions
- Reorganization handling

The maturity period must be identical across all compatible nodes.

Until finalized, miners must not assume that a newly mined reward can be spent immediately.

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13. MINING PERFORMANCE

The miner should report:

- Current hashrate
- Average hashrate
- Total hashes attempted
- Mining duration
- CPU utilization
- GPU utilization
- GPU temperature, where supported
- Accepted shares, if pool mining is enabled
- Rejected shares, if pool mining is enabled
- Blocks found
- Mining errors

Hashrate units:

Unit| Meaning
H/s| Hashes per second
kH/s| Thousand hashes per second
MH/s| Million hashes per second
GH/s| Billion hashes per second
TH/s| Trillion hashes per second

Hashrate comparisons are meaningful only when the same PoW algorithm is used.

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14. MINING SOFTWARE INTERFACE

The future SAECOIN miner should provide a simple interface.

Proposed dashboard:

```
---------------------------------
        SAECOIN MINER
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Network: TESTNET
Mode: CPU

Mining Status: RUNNING

Hashrate: 0 H/s
CPU Usage: 45%
GPU Usage: N/A

Mining Duration: 00:00:00

Blocks Found: 0

Mining Address:
Not Configured

[ START MINING ]

[ STOP MINING ]

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```

The actual interface and available metrics will depend on the final implementation.

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15. MINING CONFIGURATION

Proposed configuration:

```
[network]
network = "testnet"

[mining]
enabled = false
mode = "cpu"
threads = 4

[wallet]
mining_address = "TO_BE_CONFIGURED"

[node]
endpoint = "127.0.0.1"
port = 38445
```

This is an illustrative configuration.

The final configuration format must match the implemented miner and node interfaces.

Mining must remain disabled until a valid mining destination is configured.

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16. MINING SECURITY

Mining software must:

- Validate node responses.
- Reject malformed block templates.
- Verify submitted solutions locally where practical.
- Never expose wallet private keys.
- Avoid downloading and executing unknown mining binaries.
- Protect pool authentication credentials.
- Limit resource consumption.
- Report abnormal behavior.
- Use trusted software dependencies.

A miner must not silently change the configured payout address.

Mining software should not install hidden background services.

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17. MINING TESTING

Required tests include:

CPU Mining

- CPU detection
- Thread configuration
- Hashrate measurement
- Valid solution generation
- Invalid solution rejection
- Start/stop behavior

GPU Mining

- GPU detection
- Device selection
- Memory handling
- Hashrate measurement
- Valid solution generation
- Invalid solution rejection
- Multiple GPU testing, if supported

Solo Mining

- Node synchronization
- Block template retrieval
- Solution submission
- Block validation
- Reward accounting

Pool Mining

- Miner connection
- Job distribution
- Share validation
- Invalid share rejection
- Reward accounting
- Payout calculation

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18. MINING DECENTRALIZATION

SAECOIN aims to make mining participation accessible.

Design considerations:

- CPU mining accessibility

- GPU mining accessibility
- Reasonable hardware requirements
- Efficient verification
- Transparent mining rules
- Open-source mining software
- Independent Full Node validation

ASIC resistance is a design objective, not a guarantee.

Specialized hardware may still gain an advantage depending on the final algorithm and future hardware development.

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19. FUTURE MINING FEATURES

Potential future features:

- Graphical CPU miner
- Graphical GPU miner
- Multiple GPU support
- Mining pool dashboard
- Miner statistics
- Remote mining monitoring
- Automatic performance tuning
- Mining profitability calculator
- Mining hardware benchmark database

These features should be introduced only after the basic mining system is stable.

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20. FINAL MINING PRINCIPLES

1. All miners follow identical consensus rules.
2. Full Nodes independently validate blocks.
3. CPU mining is a project objective.
4. GPU mining is a project objective.
5. Solo mining should be implemented before pool mining.
6. Pool mining must not control consensus.
7. Mining rewards follow the finalized issuance schedule.
8. Mining software must protect wallet secrets.
9. ASIC resistance is not guaranteed.
10. Mining performance must be measured using the finalized algorithm.

DOCUMENT STATUS

Document: Mining_Guide.md
Version: 1.0
Project: SAECOIN
CPU Mining: Planned
GPU Mining: Planned
Solo Mining: Planned
Pool Mining: Future Feature
PoW Algorithm: Not Finalized
Block Reward: 50 SAC Proposed
Halving Interval: 210,000 Blocks Proposed
Implementation Status: Not Started